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Rate enhanced pulsed DC sputtering system

Abstract

A sputtering system and method are disclosed. The system includes first power source coupled between a first and second power leads, and the first power source provides a first voltage that alternates between positive and negative during each of multiple cycles. The system also includes a second power source coupled between the second power lead and a third power lead, and the second power source provides a second voltage that alternates between positive and negative during each of the multiple cycles. A controller of the system controls the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage, are simultaneously negative during a portion of each cycle and simultaneously positive during another portion of each cycle.

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Background/Summary

CLAIM OF PRIORITY UNDER 35 U.S.C. § 120 (1) The present application for patent is a continuation of U.S. patent application Ser. No. 15/802,791, entitled “Rate Enhanced Pulsed DC Sputtering System,” filed Nov. 3, 2017, which is a continuation of U.S. patent application Ser. No. 14/697,267, entitled “Rate Enhanced Pulsed DC Sputtering System,” filed Apr. 27, 2015, both of which are assigned to the assignee hereof and expressly incorporated by reference herein.

FIELD

(1) The present invention relates generally to sputtering systems, and more specifically to pulsed DC sputtering.

BACKGROUND

(2) Sputtering historically includes generating a magnetic field in a vacuum chamber and causing a plasma beam in the chamber to strike a sacrificial target, thereby causing the target to sputter (eject) material, which is then deposited as a thin film layer on a substrate, sometimes after reacting with a process gas. Sputtering sources may employ magnetrons that utilize strong electric and magnetic fields to confine charged plasma particles close to the surface of the target. An anode is generally provided to collect electrons from the plasma to maintain plasma neutrality as ions leave to bombard the target.

(3) The industry has evolved over the years in various attempts to maximize sputtering efficiency, decrease power consumption requirements, minimize the heat load of the system, minimize arcing and/or increase the types of substrates that may be used in the system.

(4) Moreover, sputtering a thin film of, for example, titanium dioxide (TiO₂) or silicon dioxide (SiO₂) onto a polyethylene substrate presents unique challenges in the industry because polyethylene is a plastic with a low melting point or low heat tolerance. Currently-available sputtering systems, whether DC or AC type systems, require a high heat load to effectuate sputtering and/or the deposition of TiO₂ or SiO₂, yet this high heat load, caused by a high current density, effectively eliminates polyethylene as a suitable substrate for many intended high power applications. Compounding the problem, if the heat load in the currently-available sputtering systems is lowered to a level that does not melt or otherwise render the polyethylene unsuitable, e.g., by reducing the power applied, the deposition rate is lowered to a point that results in a low-quality deposited layer and/or increases the required time for sputtering to a point that renders the use of polyethylene as a substrate infeasible from a commercial perspective.

(5) There therefore remains a need for a device that provides improved sputtering deposition rates at a lower heat load.

SUMMARY

(6) Embodiments disclosed herein address the above stated needs by providing a system, method, or non-transitory memory having instructions for pulsed direct current sputtering.

(7) In some aspects, a pulsed direct current system is provided. A first power source is coupled to a first and second power leads, and the first power source is configured to provide a first voltage that alternates between positive and negative relative to the first power lead during each of multiple cycles. A second power source is coupled to the second power lead and a third power lead, and the second power source is configured to provide a second voltage that alternates between positive and negative relative to the third power lead during each of the multiple cycles. A controller is configured to control the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage, are simultaneously negative during a portion of each cycle and simultaneously positive relative to the first and third power leads during another portion of each cycle.

(8) Other aspects include a non-transitory memory including non-transitory instructions. The non-transitory instructions are at least one of executable by a processor to execute a method and accessible by a field programmable gate array to configure the field programmable gate array to execute the method. The method may cause a first power source to apply a first voltage to a second power lead that alternates between positive and negative relative to a first power lead, and cause a second power source to apply a second voltage to the second power lead that alternates between positive and negative relative to a third power lead. In addition, the method includes causing the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage, are simultaneously negative and simultaneously positive at the second power lead relative to the first and third power leads over multiple cycles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram illustrating some aspects of a sputtering system;
- (2) FIG. 2 is a graph of some aspects of power applications in a sputtering system;
- (3) FIG. 3 is a block diagram illustrating aspects of some hardware components that may be implemented in the system illustrated in FIG. 1;
- (4) FIG. 4 is a flowchart of a method disclosed herein;
- (5) FIG. 5 is a graphical depiction of power supplied by an exemplary system;
- (6) FIG. 6 is a graphical depiction of heat load test results of exemplary systems described herein, compared to traditional AC systems;
- (7) FIG. 7 is a graphical depiction of a heat load resulting from the use of a system described herein at 4 kW, compared to a traditional AC system at 4 kW;
- (8) FIG. 8 is a graphical depiction of a heat load resulting from the use of a system described herein at 4 kW, compared to a traditional AC system at 8 kW; and
- (9) FIG. 9 is a graphical depiction of film thickness or deposition rates achieved using various settings of an exemplary system disclosed herein, compared to various traditional AC systems.

DETAILED DESCRIPTION

(10) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. For example, when power is applied to the first and second magnetrons **104**, **106**, the power sources **140**, **142** will generate a current and a voltage with a first polarity, and when power is applied to the anode **108**, the power sources **140**, **142** will generate a current and a voltage with a second polarity that is reversed to the first polarity. A “power cycle” is intended to reference a time period including a time of power having a voltage with the first polarity followed a time of power having a voltage with the second polarity. Further, for the purpose of this disclosure, all terms, and particularly terms such as “simultaneously” and “equal to” are intended to mean “within the tolerances of process or manufacturing controls”. For example, it will be understood that a synchronizing unit, such as unit **120**, may not achieve perfect synchronization between the power sources **140**, **142**, and therefore the term “simultaneous(ly)” is to be understood as meaning “substantially simultaneous(ly)”.

(11) Turning now to FIG. 1, an exemplary pulsed direct current sputtering system **100** is now described in detail. Generally stated, the system **100** provides a user with the ability to achieve higher deposition rates, as compared to prior AC dual magnetron and pulsed DC single magnetron sputtering approaches, by delivering higher power at a particular current. More specifically, at a particular power level, some embodiments of the system **100** disclosed herein may cut the RMS current in the endblocks or magnetrons by about half, as compared to prior AC sputtering systems. As a consequence, in cases in which the endblock current rating is limited, the system **100** as disclosed may enable delivery of nearly twice the power while staying within the endblock current rating limit.

(12) In a more specific example, the system **100** may be used by web coaters, who have in the past struggled with maintaining the sputtering process at manageable temperatures throughout production runs of over 7 days, to improve deposition rates and quality. In currently-available systems, web coaters apply ventilation to the plasma chamber **101** to reduce the heat, which increases the production rate by 35% as compared to systems without ventilation. The system **100** presently disclosed provides, in contrast, a doubling of the production rate before cooling is even applied. That is, the system **100** and/or method **400** disclosed herein may lower the heat load or heating experienced by a substrate, even under application of the same power, and may be practiced to apply a high-quality layer of material, such as silicon dioxide (SiO₂) or titanium

dioxide (TiO.sub.2), onto a polyethylene substrate, without destroying the polyethylene substrate, and in a shortened period of time as compared to those realized in currently-available systems. The SiO.sub.2 or TiO.sub.2 layer may be thicker than what may be realized in currently-available designs.

(13) The system **100** may provide about 1.5 times the deposition rate of DC single magnetron or pulsed DC single magnetron sputtering, and about 2 times the deposition rate of AC dual magnetron or bi-polar pulsed DC sputtering, with half the heat load experienced in the currently-available sputtering systems. The system **100** may include a plasma chamber **101** enclosing a first magnetron **102** engaged with a first target **103**, a second magnetron **104** engaged with a second target **105**, and an anode **108**. The system **100** may include a substrate **122** upon which the system **100** is to deposit a thin film material in a sputtering process.

(14) A first bi-polar controllable pulsed DC power supply **112** (or first bi-polar DC supply **112**) and a second bi-polar controllable pulsed DC power supply **114** (or second bi-polar DC supply **114**) may be provided. The first and second bipolar DC supplies **112**, **114** may receive direct power from a first direct current (DC) supply **116** and a second DC supply **118** respectively. The first bi-polar DC supply **112** may be coupled to the first magnetron **102** by way of power lead(s) **124**, and configured to apply a sputtering power to the first target **103**. Similarly, the second bi-polar DC supply **114** may be coupled to the second magnetron **104**, and configured to apply a sputtering power to the second target **105**. The first and second bi-polar DC supplies **112**, **114** may be coupled to the anode **108** by way of lead(s) **128**, and, more specifically, an electrical joint **129** may couple a first anode lead **113** from the first bi-polar DC supply **112** to a second anode lead **115** from the second bi-polar DC supply **114**. That is, the first power source **140** may be operatively coupled to the anode **108** and the first target **103** by way of the first magnetron **102** through leads **124**, **113**, and **128**, while the second power source **142** may be operatively coupled to the anode **108** and the second target **105** by way of the second magnetron **104** through leads **126**, **115**, and **128**.

(15) In some embodiments, the anode may be the wall of the plasma chamber **101**; however, in others and as illustrated, the anode **108** may be a floating anode **108**, and may further have a gas inlet **107** and a plurality of gas outlets **109** enabling the hollow cathode effect to aid in keeping the outlets clean for a more stable operation. Those of skill in the art will understand that, because the anode **108** is a part of the power supply delivery circuit, gas entering at the inlet **107** has an increase chance of breaking its bonds. For example, oxygen gas O.sub.2 is more likely to break into two oxygen atoms as the gas exits the plurality of outlets **109**, while nitrogen gas N.sub.2 is more likely to break into two nitrogen atoms as it exits the plurality of outlets **109**, resulting in films on the substrate **122** that are of higher quality than would be expected without the use of an anode **108** that provides a hollow cathode effect.

(16) Some embodiments provide a system **100** that maintains an anode that does not “disappear” as is known in the industry, and in the system disclosed herein, the anode **108** may be kept clean or cleansed through operation at higher temperatures, by hollow cathode discharge in the plurality of outlets **109**, and/or by sputtering directly during an application of anode power.

(17) In some embodiments, the system **100** maintains the anode **108**, which may be a floating gas anode, at a relatively high temperature suitable for maintaining a clean anode, while not causing the heat load within the chamber **101** and/or experienced by the substrate **122** to become so high as to damage the substrate **122**, such as a polyethylene substrate, therein. In some embodiments, the system **100** may maintain the anode **108** at 100° Celsius, or more. In some embodiments, the system **100** may maintain the anode **108** at a temperature of about 150° Celsius. In some embodiments, the system **100** may maintain the anode **108** at 150° Celsius or less, such as when in use by web coaters depositing titanium dioxide (TiO.sub.2) or silicon dioxide (SiO.sub.2) onto a polyethylene substrate. In some embodiments, the system **100** may maintain the anode **108** at 200° Celsius or more, such as when in use by glass coaters.

(18) Continuing with FIG. **1**, a controller **144** comprising a synchronizing unit **120** may

synchronize power signals from the first and second bi-polar DC supplies **112**, **114** in a manner that will be described in later portions of this document. In some embodiments, a first power source **140** may comprise the first bi-polar controllable pulsed DC power supply **112** and the first DC supply **116**. Similarly, a second power source may comprise the second bi-polar controllable pulsed DC power supply **114** and the second DC supply **118**.

(19) Of note, each of the first and second power supplies **140**, **142** may be arranged and configured to be aware of the other one of the first and second power supplies **140**, **142**, without attempting to control the operation of the other one of the first and second power supplies **140**, **142**. Applicant has achieved this “awareness without control” by first configuring a duty cycle (e.g. 40 kHz) of each of the first and second bi-polar DC supplies **112**, **114**, and subsequently coupling the synchronizing unit **120** and configuring one of the first and second bi-polar DC supplies **112**, **114** to be perceived as a transmitter for the purpose of frequency synchronization, and the other one of the first and second bi-polar DC supplies **112**, **114** to be perceived as a receiver, for the purpose of frequency synchronization. In contrast, each one of the first and second DC supplies **116**, **118** are independent, and do not rely on awareness of the other one of the first and second DC supplies **116**, **118** to properly function.

(20) Although not required, in one implementation, the first and second DC supplies **116**, **118** may each be realized by one or more ASCENT direct current power supplies sold by Advanced Energy Industries, Inc. of Fort Collins, Colorado, U.S.A. And the first and second bi-polar DC supplies **112**, **114** may each be realized by an ASCENT DMS Dual-magnetron sputtering accessory, which is also sold by Advanced Energy Industries, Inc. of Fort Collins, Colorado, U.S.A. In this implementation, the first and second power sources **140**, **142** are each realized as an AMS/DMS stack wherein the ASCENT direct current power supply may provide straight DC power and perform arc management functions, and the DMS dual-magnetron sputtering accessory generates a pulsed DC waveform from the straight DC power. Beneficially, the DMS dual-magnetron sputtering accessories may be located in close proximity to the chamber **101**, and the ASCENT direct current power supplies may be located remotely (e.g., in a remote rack) from the chamber **101**. The **120** synchronizing unit in this implementation may be realized by a common exciter (CEX) function of the DMS accessories.

(21) In another embodiment, each of the first and second power sources **140**, **142** may be realized by an integrated pulsed DC power supply.

(22) Turning now to FIG. 2, in some embodiments, the controller **144** and/or synchronizing unit **120** may be configured to cause the first and second bi-polar DC supplies **112**, **114** or first and second power sources **140**, **142** to simultaneously apply sputtering power to the first and second magnetrons **104**, **106**, respectively, followed by a simultaneous anode power to the anode **108**.

(23) In some embodiments, a cycle of power may be applied to the magnetrons **102**, **104** and the anode **108**. As illustrated in FIG. 2 the synchronizing unit **120** may be configured to cause the first and second bi-polar DC supplies **112**, **114** to apply power to the magnetrons **104**, **106** for a first period of time $t_{sub.1}$, followed by an application of power to the anode **108** for a second period of time $t_{sub.2}$, wherein the first period of time $t_{sub.1}$ is 80% of a sputtering cycle comprising the first and second periods of time $t_{sub.1}$, $t_{sub.2}$. The second period of time $t_{sub.2}$ may be about 20% of the cycle. In some embodiments, the first period of time $t_{sub.1}$ may be at least 70% of the cycle, or, in some embodiments, between 70% and 90% of the cycle. The second period of time $t_{sub.2}$ may be less than 30% of the cycle, or between 30% and 10% of the cycle. In some embodiments, the first period of time $t_{sub.1}$ may be between 80% and 90% of the cycle, and the second period of time $t_{sub.2}$ may be between 20% and 10% of the cycle. In some embodiments, the first period of time $t_{sub.1}$ may be between 85% and 90% of the cycle, and the second period of time $t_{sub.2}$ may be between 15% and 10% of the cycle.

(24) As discussed further herein, the controller **144** may be configured to control the first power source **140** and the second power source **142**, and may have a non-transitory memory including

non-transitory instructions to effectuate the methodologies described herein. For example, the non-transitory instructions may be accessible by a field programmable gate array to configure the field programmable gate array to execute one or more methods. In some embodiments, the non-transitory instructions are executable by a processor and/or accessible by the field programmable gate array to configure the field programmable gate array to execute one or more methods. In other embodiments, one or more aspects of the controller **144** may be realized by hardware (e.g., application specific integrated circuits) that is persistently configured to control the first and second power sources **140**, **142** to effectuate one or more of the methods described herein.

(25) Turning now to FIG. 2, certain novel and innovative aspects of a sputtering process that may be executed by the system **100** are described in detail. As illustrated in FIG. 2, the system **100** may be configured to apply a first power having a first voltage $V_{sub.1}$ to the first magnetron **102** for a first period of time $t_{sub.1}$. Simultaneously, or for the first period of time $t_{sub.1}$, the system **100** may apply a second power having a second voltage $V_{sub.2}$ to the second magnetron **104**. In some embodiments, a magnitude of the first voltage $V_{sub.1}$ may be substantially the same as a magnitude of the second voltage $V_{sub.2}$, although it will be understood that the values are highly idealized, and may not be perfectly matched in practice. The simultaneous application of the first power and the second power may result in an arbitrary summed voltage $V_{sub.sum}$ in a sputtering power applied for the first period of time $t_{sub.1}$. That is, the term “summed voltage $V_{sub.sum}$ ” is not intended to mean that the system as a whole experiences a summed voltage, but rather that the first magnetron **102** may experience a voltage and the second magnetron **104** may experience a voltage, with the voltage being summed for the purpose of analysis.

(26) Continuing with FIG. 2, the first period of time $t_{sub.1}$ may be followed by a second period of time $t_{sub.2}$. During the second period of time $t_{sub.2}$, the system **100** may be configured to cause the first power source **140** to apply an anode power having a third voltage $V_{sub.3}$ to the anode **108** and to substantially simultaneously cause the second power source **142** to apply an anode power having a fourth voltage $V_{sub.4}$ to the anode **108**. That is, the third voltage $V_{sub.3}$ and the fourth voltage $V_{sub.4}$ may be combined to apply a combined power having a combined voltage $V_{sub.combined}$ to the anode **108** for the second period of time $t_{sub.2}$, because the third anode voltage $V_{sub.3}$ and the fourth anode voltage $V_{sub.4}$ are applied together at the anode **108**. Of note, when the two power sources **140**, **142** apply power to the anode **108** simultaneously, the anode current is additive; however, the resulting combined voltage $V_{sub.combined}$ may not necessarily be, and generally is not, the sum of the third voltage $V_{sub.3}$ and the fourth voltage $V_{sub.4}$. Those of skill in the art will understand that $V_{sub.1}$, $V_{sub.2}$, and $V_{sub.combined}$ are all affected by the impedance within the chamber.

(27) While the graphical depictions in FIG. 2 are highly idealized, an approximation of the powers and voltages may be provided. In some embodiments, the first voltage $V_{sub.1}$ may be between about 300 Volts and about 800 Volts. In some embodiments, the first voltage $V_{sub.1}$ may be at least 400 Volts. A magnitude of the second voltage $V_{sub.2}$ may be substantially equal to a magnitude of the first voltage $V_{sub.1}$. A root mean square or RMS may be taken to evaluate the voltage.

(28) It should also be noted that $V_{sub.1}$ and $V_{sub.2}$ may be respectively measured relative to the anode **108**, while $V_{sub.combined}$ may be measured relative to ground. With brief reference to FIG. 5, for example, shown is an oscilloscope illustrating a top trace representing the voltage of the anode, or $V_{sub.combined}$, relative to ground, while the bottom trace illustrates a voltage applied to a first magnetron **102** relative to the anode **108**. In some embodiments, $V_{sub.combined}$ is at least 70% of $V_{sub.sum}$. In some embodiments, $V_{sub.combined}$ is at least 80% of $V_{sub.sum}$.

(29) The methods described in connection with the embodiments disclosed herein may be embodied directly in hardware, in processor executable instructions encoded in non-transitory processor readable medium, or in a combination of the two. Referring to FIG. 3 for example, shown is a block diagram depicting physical components that may be utilized to realize the

controller **144** according to an exemplary embodiment. As shown, in this embodiment a display portion **312** and nonvolatile memory **320** are coupled to a bus **322** that is also coupled to random access memory (“RAM”) **324**, a processing portion (which includes N processing components) **326**, a field programmable gate array (FPGA) **327**, and a transceiver component **328** that includes N transceivers. Although the components depicted in FIG. 3 represent physical components, FIG. 3 is not intended to be a detailed hardware diagram; thus many of the components depicted in FIG. 3 may be realized by common constructs or distributed among additional physical components. Moreover, it is contemplated that other existing and yet-to-be developed physical components and architectures may be utilized to implement the functional components described with reference to FIG. 3.

(30) This display portion **312** generally operates to provide a user interface for a user, and in several implementations, the display is realized by a touchscreen display. In general, the nonvolatile memory **320** is non-transitory memory that functions to store (e.g., persistently store) data and processor executable code (including executable code that is associated with effectuating the methods described herein). In some embodiments for example, the nonvolatile memory **320** includes bootloader code, operating system code, file system code, and non-transitory processor-executable code to facilitate the execution of a method described with reference to FIG. 4.

(31) In many implementations, the nonvolatile memory **320** is realized by flash memory (e.g., NAND or ONENAND memory), but it is contemplated that other memory types may be utilized as well. Although it may be possible to execute the code from the nonvolatile memory **320**, the executable code in the nonvolatile memory is typically loaded into RAM **324** and executed by one or more of the N processing components in the processing portion **326**.

(32) The N processing components in connection with RAM **324** generally operate to execute the instructions stored in nonvolatile memory **320** to enable the power sources **140**, **142** to achieve one or more objectives. For example, non-transitory processor-executable instructions to effectuate the methods described with reference to FIG. 4 may be persistently stored in nonvolatile memory **320** and executed by the N processing components in connection with RAM **324**. As one of ordinary skill in the art will appreciate, the processing portion **326** may include a video processor, digital signal processor (DSP), graphics processing unit (GPU), and other processing components.

(33) In addition, or in the alternative, the FPGA **327** may be configured to effectuate one or more aspects of the methodologies described herein (e.g., the method described with reference to FIG. 4). For example, non-transitory FPGA-configuration-instructions may be persistently stored in nonvolatile memory **320** and accessed by the FPGA **327** (e.g., during boot up) to configure the FPGA **327** to effectuate the functions of the controller **144**.

(34) The input component operates to receive signals that are indicative of one or more aspects of the power applied to the first magnetron **102** and/or the second magnetron **104**. The signals received at the input component may include, for example, voltage, current, and/or power. The output component generally operates to provide one or more analog or digital signals to effectuate an operational aspect of the first and/or second power sources **140**, **142**. For example, the output portion may be a signal to cause the first power source and/or second power source **112**, **114** to effectuate some of the methodologies described with reference to FIG. 4. In some embodiments, the output component operates to adjust a frequency and duty of the first and/or second power source **140**, **142**.

(35) The depicted transceiver component **328** includes N transceiver chains, which may be used for communicating with external devices via wireless or wireline networks. Each of the N transceiver chains may represent a transceiver associated with a particular communication scheme (e.g., WiFi, Ethernet, Profibus, etc.).

(36) Turning now to FIG. 4, a method **400** of sputtering that may be executed by the system **100** in some embodiments is now described in more detail. The method **400** may include providing **402** a plasma chamber, a first power source, a second power source, and other sputtering components.

Providing **402** may be achieved by providing a system such as system **100** described with reference to FIGS. **1-3**. The method **400** may include causing **404** the first power source to apply a first sputtering power to the first magnetron for a first period of time. The method **400** may also include causing **406** the second power source to apply a second sputtering power to the second magnetron for the first period of time, the first sputtering power and the second sputtering power providing a summed sputtering voltage. In some embodiments, the method **400** may include calculating **414** one or more of a summed sputtering power, a summed sputtering voltage, and a summed sputtering current applied to the first magnetron and the second magnetron.

(37) The method **400** may also include causing **408** the first power source to apply a first anode power to the anode for a second period of time following the first period of time. The method **400** may also include causing **410** the second power source to apply a second anode power to the anode for the second period of time, the first anode power and the second anode power providing a combined anode power.

(38) The method **400** may include combining **412** an anode power to the anode such that the combined anode power, having a current and a voltage, has a voltage having a magnitude of at least 80% of a magnitude of the summed sputtering voltage.

(39) In method **400**, the first period of time may be at least 80% of a sputtering cycle, the sputtering cycle comprised of the first period of time and the second period of time. In some embodiments, the first period of time is at least 70% of the sputtering cycle. In some embodiments, the combined anode voltage has a magnitude of at least 800 Volts. In some embodiments, the first period of time $t_{sub.1}$ may be between 70% and 90% of the cycle, and the second period of time $t_{sub.2}$ may be between 30% and 10% of the cycle. In some embodiments, the first period of time $t_{sub.1}$ may be between 80% and 90% of the cycle, and the second period of time $t_{sub.2}$ may be between 20% and 10% of the cycle. In some embodiments, the first period of time $t_{sub.1}$ may be between 85% and 90% of the cycle, and the second period of time $t_{sub.2}$ may be between 15% and 10% of the cycle.

(40) In some embodiments, the first sputtering power has a voltage having a magnitude of at least 300 Volts, and the second sputtering power has a voltage having a magnitude of at least 300 Volts.

(41) In some embodiments, the combined anode power has a current and a voltage, the voltage having a magnitude of at least 1000 Volts.

(42) In some embodiments, the anode is a floating anode comprising a gas bar having a gas inlet and a plurality of gas outlets shaped to provide a Hollow Cathode Effect.

(43) In some embodiments, the method **400** includes causing a synchronizer to cause the first power source and the second power source to simultaneously apply power to the first magnetron and the second magnetron for the first period of time and to cause the first power source and the second power source to simultaneously apply power to the anode for the second period of time.

(44) To determine basic functionality, the system **100** and method **400** previously described herein were tested using $TiO_{sub.x}$ as the target material, 6.4 mTorr using 126 SCCM argon and 100 SCCM oxygen as the reactive gas, 4 kW of applied power per magnetron with a floating anode sitting between them, and a line speed of 10 inches per minute. Table 1 illustrates the results of the functionality test, establishing that the system **100** would operate.

(45) TABLE-US-00001 TABLE 1 Functionality Test Observations at first plasma. Item Voltage Current V1 (V of first magnetron 102 to anode 108) -535 V 7.2 A V2 (V of second magnetron 104 to anode -565 V 7.1 A 108) Voltage of Anode 108 to Ground 850 V NA Anode -680 V.sub.combined 2.2 A Voltage of first magnetron 102 to Ground -720 V NA

(46) The above functionality test verified that the system **100** was functional substantially as envisioned, and was attempting to deliver full power to the anode **108**. Of note, an unforeseen result of the functionality test revealed that the anode power exhibited a current of just 2.2 Amperes, or about half of what was expected.

(47) Applicant has determined that, because there was no magnetic enhancement on the anode **108**, it would take very high voltages to do any more work beyond heating the anode **108**, which was

also found to be a desirable response. That is, V.sub.combined is high enough to result in a very low current at the anode **108** in the functionality test. Moreover, because the current is additive, a current at the anode **108** of just 2.2 Amperes indicates that a current approaching 0 is experienced at each of the first and second magnetrons **102, 104**—another desirable result to prevent undesirable coating of the first and second magnetrons **102, 104** during an anode power pulse. (48) Beyond the functionality test, the system was tested under three other settings, described below under runs 1, 2, and 3. Benchmark runs 4 and 5 were also performed using a traditional AC sputtering system as described below. By measuring a film deposition thickness at the same given line speed, deposition rates could be calculated. The heat load or temperature of the substrate **122** was also measured, as was the voltage and current at the magnetrons **102, 140** and anode **108**. Of note, the setup was run on a sputter-down machine with relatively small magnetrons, so process power was limited.

(49) Turning now to FIG. **6**, illustrating a summary of the heat load responses, a series of tests were performed at various frequencies, notably run 1 was at 20 kHz, run 2 was at 30 kHz, and run 3 was at 40 kHz with as much Side A ON time as possible, that is, with as much time of power application to the magnetrons.

(50) To compare deposition rates and heat loads, the system **100** was compared to a currently available Advanced Energy PEII AC power supply, illustrated at benchmark test runs 4 and 5.

(51) With each run, a film thickness measurement was taken using a Dektak® profilometer, with a Sharpie® mark on the bare glass. Scrubbing the coated glass over the Sharpie mark removes the coating so a good thickness step could be obtained.

(52) Also with each run, the heat load on the substrate was measured by a SuperMole, which is a circuit board encased by many heat shields with a type K thermocouple super-glued to the glass substrate. The SuperMole took real-time temperature measurements through the plasma and the results were then downloaded and stored.

(53) With simultaneous reference now to FIGS. **6-9**, the 5 runs were completed, with all machine setups maintained as described above. The results of the 5 runs are described below.

(54) TABLE-US-00002 TABLE 2 Run 1. System 100 @ 4 kW, 20 kHz. DC Supply Settings Power Voltage Current 1.sup.st DC supply 116 Settings 4 kW V.sub.1 = 513 V I.sub.1 = 7.8 A 2.sup.nd DC supply 118 Settings 4 kW V.sub.2 = 526 V I.sub.2 = 7.7 A Side A Settings Frequency Side A Side B Boost 1.sup.st bi-polar DC supply 112 20 kHz 90% 10% 50% 2.sup.nd bi-polar DC supply 114 Side B (Anode 108) Settings Voltage Current 1.sup.st bi-polar DC supply 112 V.sub.3 = 589 V I.sub.3 = 2.8 A 2.sup.nd bi-polar DC supply 114 V.sub.4 = 598 V I.sub.4 = 3.4 A Anode V.sub.combined not I.sub.combined = 6.2 A measured

(55) TABLE-US-00003 TABLE 3 Run 2. System 100 @ 4 kW, 30 kHz. DC Supply Settings Power Voltage Current 1.sup.st DC supply 116 Settings 4 kW V.sub.1 = 527 V I.sub.1 = 7.6 A 2.sup.nd DC supply 118 Settings 4 kW V.sub.2 = 542 V I.sub.2 = 7.4 A Side A Settings Frequency Side A Side B Boost 1.sup.st bi-polar DC supply 112 30 kHz 85% 15% 50% 2.sup.nd bi-polar DC supply 114 Side B (Anode 108) Settings Voltage Current 1.sup.st bi-polar DC supply 112 V.sub.3 = 637 V I.sub.3 = 2.2 A 2.sup.nd bi-polar DC supply 114 V.sub.4 = 647 V I.sub.4 = 2.2 A Anode V.sub.combined not I.sub.combined = 4.4 A measured

(56) TABLE-US-00004 TABLE 4 Run 3. System 100 @ 4 kW, 40 kHz. DC Supply Settings Power Voltage Current 1.sup.st DC supply 116 Settings 4 kW V.sub.1 = 545 V I.sub.1 = 7.3 A 2.sup.nd DC supply 118 Settings 4 kW V.sub.2 = 552 V I.sub.2 = 7.2 A Side A Settings Frequency Side A Side B Boost 1.sup.st bi-polar DC supply 112 40 kHz 80% 20% 50% 2.sup.nd bi-polar DC supply 114 Side B (Anode 108) Settings Voltage Current 1.sup.st bi-polar DC supply 112 V.sub.3 = 678 V I.sub.3 = 2.2 A 2.sup.nd bi-polar DC supply 114 V.sub.4 = 680 V I.sub.4 = 2.2 A Anode V.sub.combined not I.sub.combined = 4.4 A measured

(57) TABLE-US-00005 TABLE 5 Run 4. Traditional AC System @ 4 kW. PEII Settings Power Voltage Current PEII AC Power Settings 4 kW 570 V 7.6 A

(58) TABLE-US-00006 TABLE 6 Run 5. Traditional AC System @ 8 kW. PEII Settings Power Voltage Current PEII AC Power Settings 8 kW 630 V 14 A

(59) FIG. 7 illustrates the heat load results of the system **100** running at 40 kHz and 4 kW, as compared to an AC system at 4 kW. FIG. 8 illustrates the heat load results of the system **100** running at 40 kHz and 4 kW, as compared to an AC system at 8 kW. FIG. 9 illustrates the resulting film thickness for each of the 5 runs (runs 1-3 being test runs and runs 4-5 being benchmark tests of AC systems).

(60) With reference now to FIGS. 7 and 9, FIG. 7 indicates that the system **100** results in less heating (under 300 degrees Celsius versus over 300 degrees Celsius). Moreover, and with reference to tables 3 and 5, the system **100** provides a combined current $I_{combined}$ of 4.4 Amperes at a power of 4 kW, while an AC system at 4 kW provides a current of 7.6 Amperes. That is, at the same power wattage settings, the system **100** causes the anode **108** to experience significantly less current (4.4 Amperes versus 7.6 Amperes), resulting in a relatively hot anode, improving cleaning properties of the anode, and yet maintaining a relatively cool substrate. In light of FIG. 9, the system **100** at the same power settings of a traditional AC system results in a much improved deposition rate (over 350 Angstroms versus about 225 Angstroms), meaning that a processing facility may realize significant yields without damage to sensitive substrates, such as polyethylene.

(61) With reference now to FIGS. 8 and 9, illustrating the system **100** with the first and second bipolar DC supplies **112**, **114** operating at 40 kHz, and the first and second DC supplies **116**, operating at 4 kW per side, compared to a traditional PEII AC system running at 8 kW, it is noted that, although the traditional system resulted in a thicker film, the system **100** resulted in a much lower heat load. That is, although the deposition rate was just slightly more with the traditional system, the heat load was half on the system **100** as herein described. While this is interesting it is not where a usual process engineer would be when an AC power or current limit is hit and more deposition rate is required. For example, if a sputtering zone is running at 90 kW and is hitting the 300 amp AC current limit a traditional system, a system **100** is described herein could be used instead. Placing two 120 kw power supplies **140**, **142** in place of a 120 kW AC delivery system would result in the dramatic differences illustrated in FIGS. 7 and 9. With a system **100** as herein described, it is possible to push more power into the power-delivery system without breaching the inductive heating current limit. The system **100** may improve production speeds, add deposition rates without adding cathode lids, use existing cathodes to their fullest extent, and keep substrates much cooler.

(62) With reference to tables 3 and 6, one can see that the system **100** provides a combined current $I_{sub.combined}$ of 4.4 Amperes at a power of 4 kW, while an AC system at 8 kW provides a current of 14 Amperes. Taken in light of FIG. 9, one can see that the system **100** at 4 kW and 40 kHz results in a deposition rate that approaches that of a traditional AC system at twice the power, but resulting in significantly less current (4.4 Amperes versus 14 Amperes) and a much lower heat load (about 300 degrees Celsius versus over 500 degrees Celsius).

(63) A close analysis of FIG. 9 reveals that using a first power supply **140** and a second power supply **142** as previously described herein increases the deposition rates as compared to a traditional AC type sputtering system at the same power. For example, the system **100** running at 4 kW per side and 20 kHz results in a film thickness of 470 Angstroms, while the traditional PEII AC system at 4 kW results in a film thickness of just 230 Angstroms. To approach the deposition rates of the system **100**, a traditional AC system would have to apply more than 8 kW of power.

(64) Put succinctly, Applicant has provided a system, method, and means for increasing the deposition rate while drastically reducing the heat load experienced by the substrate **122** at the same or lower power as that of a traditional AC system, and the current experienced by the anode **108** and magnetrons **102**, **104**, which, as previously described, was an unexpected result.

(65) Returning now to FIG. 6, one can deduce that system **100** may run at any number of frequencies, including 20, 30, and 40 kHz, or others and provide a much better Angstrom/substrate

temperature profile consistently over traditional AC power delivery.

(66) The previous description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the present invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A pulsed power system, comprising: a first power lead, a second power lead, and a third power lead; a first power source coupled to the first power lead and the second power lead, the first power source configured to provide a first voltage to the first power lead that alternates between positive and negative relative to the second power lead during each of multiple cycles; a second power source coupled to the second power lead and the third power lead, the second power source configured to provide a second voltage to the third power lead that alternates between positive and negative relative to the second power lead during each of the multiple cycles; and a controller programmed to control the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage simultaneously cross zero volts.
2. The system of claim 1, wherein: the controller comprises a processor, and the controller is configured with a non-transitory memory comprising non-transitory instructions executable by the processor.
3. The system of claim 1, wherein: the controller comprises a field programmable gate array, and the controller is configured with a non-transitory memory including non-transitory instructions accessible by the field programmable gate array to configure the field programmable gate array.
4. The system of claim 1, wherein: the first power source comprises a first direct current power supply coupled to a first bi-polar controllable pulsed direct current power supply, the first bi-polar controllable pulsed direct current power supply configured to provide an alternating direct current power to the first power lead and the second power lead; and the second power source comprises a second direct current power supply coupled to a second bi-polar controllable pulsed direct current power supply, the second bi-polar controllable pulsed direct current power supply configured to provide an alternating direct current power to the third power lead and the second power lead.
5. The system of claim 1, wherein: the controller is configured to control the first power source and the second power source, so both, the first voltage and the second voltage are simultaneously positive relative to the second power lead at least 70 percent of the multiple cycles.
6. A non-transitory memory including non-transitory instructions that are, at least one of, executable by a processor to execute a method and accessible by a field programmable gate array to configure the field programmable gate array to execute the method, the method comprising: causing a first power source to apply a first voltage to a first power lead that alternates between positive and negative relative to a second power lead; causing a second power source to apply a second voltage to a third power lead that alternates between positive and negative relative to the second power lead; and causing the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage simultaneously cross zero volts.
7. The non-transitory memory of claim 6, wherein: the first voltage and the second voltage are simultaneously positive at least 70 percent of a time over the multiple cycles.
8. The non-transitory memory of claim 6, wherein: the first voltage has a positive magnitude that is the same as a negative magnitude of the first voltage; and the second voltage has a positive magnitude that is the same as a negative magnitude of the second voltage.

9. A method of pulsed direct current sputtering, comprising: providing a first power lead, a second power lead, and a third power lead; coupling a first power source to the first power lead and the second power lead; coupling a second power source to the third power lead and the second power lead; coupling a controller to both the first and second power sources; applying, with the first power source, a first voltage to the first power lead that alternates between positive and negative relative to the first-second power lead; applying, with the second power source, a second voltage to the third power lead that alternates between positive and negative relative to the second power lead; and phase-synchronizing, with the controller, the first voltage with the second voltage, so both, the first voltage and the second voltage, are simultaneously cross zero volts.
10. The method of claim 9, wherein: the first voltage and the second voltage at the second power lead are simultaneously positive relative to the second power lead at least 70 percent of a time over the multiple cycles.
11. The method of claim 10, wherein: the first voltage and the second voltage are simultaneously positive relative to the second power lead at least 80 percent of the time over the multiple cycles.
12. The method of claim 11, wherein: the first voltage and the second voltage are simultaneously positive relative to the second power lead between 70 and 90 percent of a time over the multiple cycles.
13. The method of claim 9, wherein: the first voltage has a positive magnitude that is the same as a negative magnitude of the first voltage; and the second voltage has a positive magnitude that is the same as a negative magnitude of the second voltage.
14. The method of claim 9 including: providing the second power lead as a floating power lead.
15. The system of claim 1, wherein the controller is programmed to control the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage simultaneously cross zero volts two times in each cycle.
16. The system of claim 1, wherein the controller is programmed with hardware.
17. The system of claim 6, wherein the non-transitory memory comprises non-transitory instructions to cause the first power source and the second power source to phase-synchronize the first voltage with the second voltage, so both, the first voltage and the second voltage simultaneously cross zero volts two times in a cycle.
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